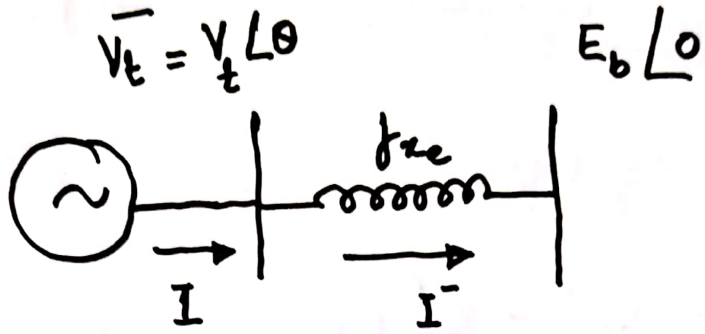


$$\bar{V}_t = jx_e \bar{I} + E_b$$

$$(V_d + jV_q)\, e^{j(\delta - \pi/2)} = jx_e (I_d + jI_q)\, e^{j(\delta - \pi/2)} + E_b$$

$$\Rightarrow \quad V_d + jV_q = jx_e (I_d + jI_q) + E_b\, e^{-j(\delta - \pi/2)}$$

$$V_d + jV_q = jx_e I_d - x_e I_q + E_b \left[\cos(\delta - \pi/2) - j\sin(\delta - \pi/2)\right]$$

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$$V_d + jV_q = jX_e I_d - X_e I_q + E_b \left[\cos\delta \cos\tfrac{x}{2} + j \sin\delta \sin\tfrac{x}{2} - j\left(\sin\delta \cos\tfrac{x}{2} - \sin\tfrac{x}{2} \cos\delta \right) \right]$$

$$V_d + jV_q = jX_e I_d - X_e I_q + E_b \sin\delta + jE_b \cos\delta$$

$$V_d = -X_e I_q + E_b \sin\delta$$

$$X_q I_q = -X_e I_q + E_b \sin\delta$$

$$V_d = X_q I_q$$

$$I_q = \frac{E_b \sin\delta}{X_q + X_e} \quad \text{— (3)}$$

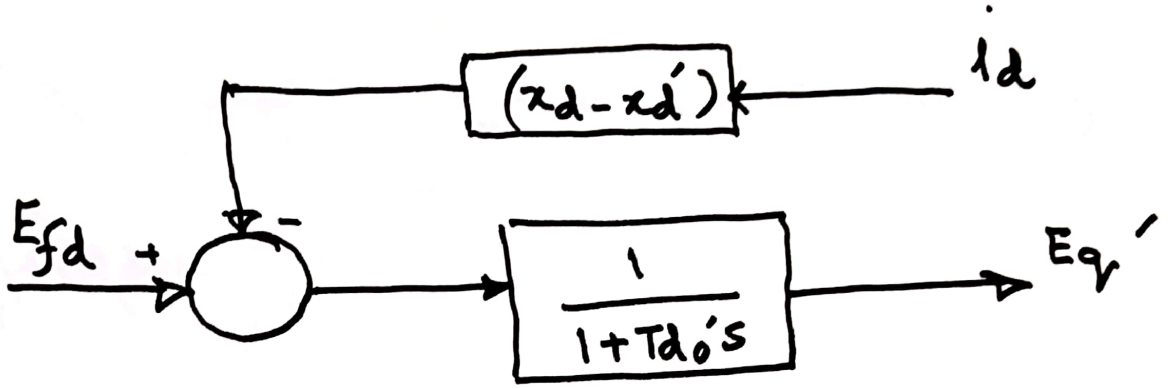
$$V_q = X_e I_d + E_b \cos\delta$$

$$V_q = E_q' - X_d' I_d$$

$$E_q' - X_d' I_d = X_e I_d + E_b \cos\delta$$

$$\Rightarrow I_d = \frac{E_q' - E_b \cos\delta}{X_e + X_d'} \quad \text{— (2)}$$

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$$E_q' = \left(\frac{1}{1+T_{do}'s}\right)\left[E_{fd} - (x_d - x_d')i_d\right]$$

$$E_q' + T_{do}'E_q'^{\cdot} = E_{fd} - (x_d - x_d')i_d$$

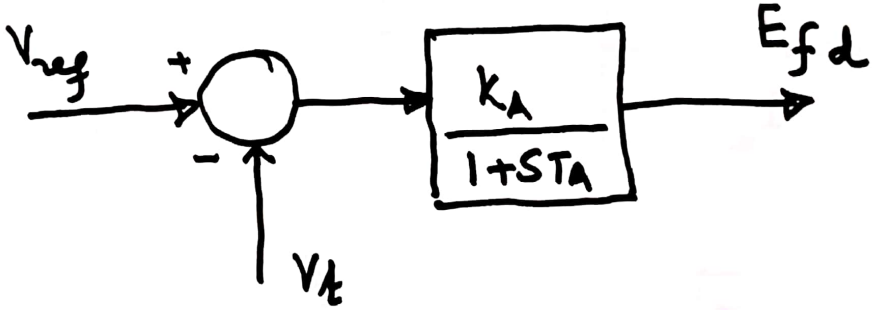
$$E_q'^{\cdot} = \frac{E_{fd} - (x_d - x_d')i_d - E_q'}{T_{do}'} \quad \text{— (4)}$$

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$$T_e = V_d i_d + V_q i_q$$

$$= x_q i_d i_q + (E_q' - x_d' i_d) i_q$$

$$= (x_q - x_d') i_d i_q + E_q' i_q \quad \text{— (5)}$$



$$E_{fd} = \left(\frac{K_A}{1+ST_A}\right)(V_{ref} - V_t)$$

$$E_{fd} + T_A \dot{E}_{fd} = K_A (V_{ref} - V_t)$$

$$\dot{E}_{fd} = \frac{K_A(V_{ref} - V_t) - E_{fd}}{T_A} \quad \text{— (8)}$$

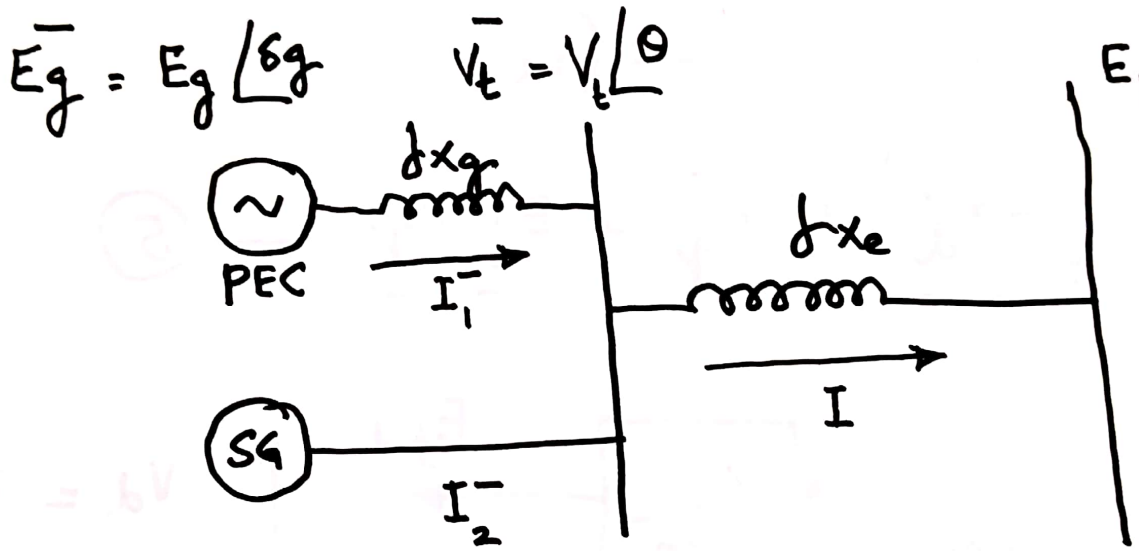
$$V_d = x_q i_q \quad \text{— (7)}$$

$$V_q = E_q' - x_d' i_d \quad \text{— (8)}$$

$$V_t = \sqrt{V_d^2 + V_q^2} \quad \text{— (9)}$$

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SMIB System with Grid-forming Converter (PEC)



$$\bar{I} = \bar{I_1} + \bar{I_2}$$

$$= \bar{I_1} + \frac{\bar{E_g} - \bar{V_t}}{jx_g} \quad \text{—— (1)}$$

Now

$$\bar{V}_t = E_b + jx_e\left(\bar{I}_1 + \frac{E_g e^{j\delta_g} - V_t e^{j\theta}}{jx_g}\right) \quad \text{—— (2)}$$

Now $\bar{V}_t = (V_d + jV_q)\, e^{j(\delta - \pi/2)}$

$$\bar{I}_1 = (I_d + jI_q)\, e^{j(\delta - \pi/2)}$$

$\therefore$ (2) can be written as:

$$(V_d + jV_q)\, e^{j(\delta-\pi/2)} = E_b + jx_e\left[(I_d + jI_q)\, e^{j(\delta+\pi/2)}\right] + \frac{E_g x_e}{x_g} e^{j\delta_g} - \frac{V_t x_e}{x_g} e^{j\theta}$$

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Multiply both sides by $e^{-j(\delta - \pi/2)}$ we get

$$V_d + jV_q = E_b \sin\delta + jE_b \cos\delta + j(x_e)(I_d + jI_q) + \frac{x_e}{x_g} E_g e^{j(\delta_g - \delta + \pi/2)} - \frac{x_e}{x_g} V_t e^{j(\theta - \delta + \pi/2)}$$

$$= \left[E_b \sin\delta + jE_b \cos\delta\right] + \left[-x_e I_q + jx_e I_d\right] + \frac{x_e E_g}{x_g}\left[\sin(\delta_g - \delta)\right] + j\frac{x_e E_g}{x_g}\left[\cos(\delta_g - \delta)\right]$$

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$$+ \frac{x_e}{x_g} V_t \sin(\theta - \delta)$$

$$- \frac{x_e}{x_g}(V_d + jV_q)$$

Multiply both sides by x_g we have

$$x_g V_d + j x_g V_q = x_g(E_b \sin\delta + j E_b \cos\delta) + x_g(-x_e I_q + j x_e I_d)$$

$$- x_e E_g \sin(\delta_g - \delta) + j x_e E_g \cos(\delta_g - \delta)$$

$$- x_e V_d + j x_e V_q$$

Comparing real parts

$$(x_g + x_e)V_d = x_g E_b \sin\delta - x_g x_e I_q - x_e E_g \sin(\delta_g - \delta)$$

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but $v_d = x_q i_q$

$$\therefore \left[\begin{matrix} x_q(x_q + x_e) \\ + x_q x_e \end{matrix}\right] i_q = x_q E_b \sin\delta - x_e E_g \sin(\delta_g - \delta)$$

$$\Rightarrow \boxed{i_q = \frac{x_q E_b \sin\delta - x_e E_g \sin(\delta_g - \delta)}{x_q x_q + x_q x_e + x_g x_e}}$$

Now comparing imaginary parts

$$x_q v_q' = x_q E_b \cos\delta + x_q x_e i_d + x_e E_g \cos(\delta_g - \delta) - x_e v_q$$

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$$(X_g + X_e) V_q = X_g E_b \cos\delta + X_g X_e i_d - X_e E_g \cos(\delta_g - \delta)$$

but $V_q = E_q' - X_d' i_d$

$$\therefore (X_g + X_e)(E_q' - X_d' i_d) = X_g E_b \cos\delta + X_g X_e i_d + X_e E_g \cos(\delta_g - \delta)$$

$$X_g E_q' - X_g X_d' i_d + X_e E_q' - X_e X_d' i_d$$

$$= X_g E_b \cos\delta + X_g X_e i_d + X_e E_g \cos(\delta_g - \delta)$$

$$[X_g X_e + X_g X_d' + X_e X_d'] i_d = (X_g + X_e) E_q' - X_g E_b \cos\delta - X_e E_g \cos(\delta_g - \delta)$$

$$\therefore \boxed{i_d = \frac{(X_g + X_e) E_q' - X_g E_b \cos\delta - X_e E_g \cos(\delta_g - \delta)}{X_g X_e + X_g X_d' + X_e X_d'}}$$

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